

Geometry-Optimized Design and Experimental Validation of a Compact 2.4 GHz Microstrip Patch Antenna for RF Energy Harvesting

Hüseyin Aslan ^{a,1}, Davut Izci ^{a,2,*}, Hüseyin Özmen ^{b,3}, Serdar Ekinci ^{c,4}, Alfian Ma'arif ^{d,5}

^a Department of Electric and Electronic Engineering, Bursa Uludag University, Bursa, Turkey

^b Department of Electric & Electronic Engineering, Dicle University, Diyarbakır, Turkey

^c Department of Computer Engineering, Bitlis Eren University, Bitlis, Turkey

^d Department of Electrical Engineering, Universitas Ahmad Dahlan, Yogyakarta, Indonesia

¹ 12505004@ogr.uludag.edu.tr; ² davutizci@uludag.edu.tr; ³ huseyin.ozmen@dicle.edu.tr; ⁴ sekinici@beu.edu.tr;

⁵ alfianmaarif@ee.uad.ac.id

* Corresponding Author

ARTICLE INFO

Article history

Received May 13, 2026

Revised June 04, 2026

Accepted June 11, 2026

Keywords

RF Energy Harvesting;
Microstrip Patch Antenna;
Single-Band Antenna;
Wireless Power Transfer;
Antenna Optimization;
Robotic Applications

ABSTRACT

The rapid expansion of wireless infrastructures has intensified the demand for energy-autonomous electronic systems capable of operating without conventional power supplies. In this context, Radio Frequency (RF) energy harvesting has emerged as a viable solution for powering low-consumption devices by exploiting ambient electromagnetic radiation. In this study, two single-band microstrip patch antennas operating at 2.4 GHz are designed for RF energy harvesting applications. The initial dimensions are obtained using the transmission-line model, after which the radiating surfaces are refined through geometry-driven modifications involving unique slots and notches. These structural adaptations are introduced to reshape surface current paths, extend the effective electrical length, and enhance impedance bandwidth and radiation characteristics. Extensive full-wave simulations are conducted in CST Microwave Studio, and hundreds of parametric iterations are performed to identify optimal configurations. The antenna exhibiting the best electromagnetic performance is fabricated on an FR-4 substrate using standard printed circuit board technology. Experimental validation is carried out using a vector network analyzer, revealing close agreement between simulated and measured results. The fabricated antenna achieves a resonance at 2.4 GHz with a VSWR of 1.0, a reflection coefficient of -31.01 dB, a gain of 2.75 dBi, and a directivity of 5.08 dBi. When integrated with a commercial RF energy harvesting module, the antenna is experimentally demonstrated to deliver a DC voltage of 3.51 V under operating conditions. The results confirm that the proposed compact and low-cost antenna provides an effective balance between miniaturization, impedance matching, and radiation efficiency. By combining analytical formulation, geometry-driven refinement, and experimental verification, this work presents a practical and reproducible approach for high-performance antennas in RF energy harvesting and wireless power transfer systems.

© 2025 The Authors.

Published by Association for Scientific Computing Electrical and Engineering.

This is an open-access article under the [CC-BY-NC](https://creativecommons.org/licenses/by-nc/4.0/) license.



1. Introduction

Microstrip patch antennas were conceptually introduced in 1953, patented in 1955, and began to gain widespread adoption during the 1970s. Since then, they have become one of the most prominent radiating structures in modern communication systems owing to their low fabrication cost, planar geometry, mechanical robustness, and compatibility with printed circuit technologies [1]–[3]. Their compact size and lightweight nature have enabled extensive deployment in a wide range of applications, including aircraft and space technologies, wireless energy systems, multiple-input multiple-output (MIMO) platforms, the internet of things (IoT), and radar systems [4]–[9].

In parallel with the rapid growth of wireless infrastructures, the demand for sustainable and autonomous energy solutions has intensified. Radio frequency (RF) energy harvesting has emerged as a promising approach for supplying low-power electronic devices by exploiting ambient electromagnetic radiation. In this context, microstrip patch antennas have gained particular importance because of their ability to efficiently capture RF signals even at low power densities, their ease of integration with rectification circuits, and their low manufacturing cost. Their small profile, planar structure, and economic feasibility [10], together with their suitability for smart grids, WLAN and WiMAX systems [8], military platforms, and autonomous sensor networks, have rendered them indispensable components of modern wireless ecosystems [11]–[16].

A substantial body of literature has focused on enhancing the performance of microstrip antennas through geometric modification and optimization. Sahoo et al. [17] proposed a Z-shaped antenna optimized via discrete particle swarm optimization for WiMAX and S-band applications, operating in the 3.5–4.3 GHz range. Dey et al. [18] designed a rectangular microstrip antenna operating at 3 GHz and 18 GHz and later refined its resonance using particle swarm optimization. Manoj and Rodriguez [19] employed genetic algorithms to improve bandwidth and return loss for a 5 GHz microstrip antenna. Motin et al. [20] introduced slotted antennas using a dielectric substrate with $\epsilon_r = 2.2$ and a thickness of 1.588 mm, achieving favorable reflection coefficients across X-, Ku-, and K-bands. Özmen and Kurt [21] developed an FR-4 based antipodal Vivaldi antenna for ultra-wideband microwave imaging, operating over 3.05–12.2 GHz.

Further contributions have explored compactness and multi-band functionality. Imtiaj and Bhattacharya [22] compared 2×1 and 2×2 antenna arrays with respect to radiation and impedance characteristics. Jayasinghe et al. [23] applied genetic algorithms to irregularly overlapped patch cells, enabling operation over GSM1800, GSM1900, UMTS, LTE2300, and Bluetooth bands. Rasheda et al. [24] developed UWB-compliant antennas covering 3.1–10.6 GHz and demonstrated improved gain and VSWR at 3.5 GHz and 6 GHz. Zhang et al. [25] introduced wide-angle antennas for RF energy harvesting and wireless power transfer, covering 2G–5G and ISM bands, with measured harvested power levels of -8.5 dBm and -8.9 dBm at 2.4 GHz and 5.8 GHz, respectively.

Additional efforts have emphasized gain enhancement and size reduction. Orankitanun and Yaowiwat [26] achieved gains exceeding 8 dBi across multiple bands for WCDMA and WiMAX applications. Boudjerda et al. [27] reduced resonance frequency without altering patch dimensions by introducing thin-film layers, enabling antenna miniaturization. Islam et al. [28] improved gain and bandwidth in microwave imaging systems through rectangular slot integration, while Mollah et al. [29] demonstrated graphene-based microstrip antennas with gains above 6.8 dBi and reflection coefficients below -23 dB.

Despite these advances, several challenges remain evident in the literature, particularly for RF energy harvesting applications. Many existing designs exhibit limited miniaturization, moderate gain, or insufficient suppression of reflected power, all of which directly constrain the efficiency of the harvesting chain. Moreover, a considerable portion of reported studies remains at the simulation stage, with limited experimental validation under realistic operating conditions [30]–[34].

Motivated by these gaps, the present study proposes a compact, single-band microstrip patch antenna specifically tailored for RF energy harvesting in the 2.4 GHz ISM band. The antenna geometry is initially derived using analytical expressions based on the transmission-line model and

subsequently refined through extensive full-wave simulations in CST Microwave Studio, a platform widely adopted in microwave and defense-oriented electromagnetic design. During the optimization process, unique slots and notches are introduced on the radiating surface to reshape current paths, extend the effective electrical length, and enhance impedance bandwidth and directivity [35], [36]. Unlike many existing approaches that rely primarily on algorithmic optimization, the present work adopts a geometry-driven refinement strategy grounded in electromagnetic insight.

Two antenna configurations are developed and evaluated. The first serves as a reference structure, while the second incorporates further geometric compaction and refinement. Hundreds of simulations are performed to identify optimal dimensional combinations. One of the optimized designs is fabricated on an FR-4 substrate ($\epsilon_r = 4.4$) using standard PCB technology, ensuring practical feasibility and low production cost.

Experimental validation is carried out using a vector network analyzer, and the measured impedance characteristics are shown to closely match the simulation results. The first antenna exhibits a resonance at 2.4 GHz with a VSWR of 1.0, a reflection coefficient of -29.74 dB, a gain of 2.67 dBi, and a directivity of 6.6 dBi. The fabricated second antenna achieves a VSWR of 1.0, an S_{11} value of -31.01 dB, a gain of 2.75 dBi, and a directivity of 5.08 dBi. When integrated with a commercial RF energy harvesting module, the proposed antenna is experimentally demonstrated to deliver a DC voltage of 3.54 V. These results confirm that the proposed compact microstrip patch antenna provides a balanced combination of miniaturization, impedance matching, and radiation efficiency. By coupling analytical formulation, geometry-driven refinement, and experimental verification, this work introduces a practical and reproducible pathway toward high-performance antennas for RF energy harvesting and wireless power transfer systems.

2. Antenna Parameters

The electromagnetic performance of a microstrip patch antenna is governed by a set of interrelated parameters that determine its radiation behavior, impedance matching, and suitability for energy harvesting. In this study, the antenna geometry is derived using the transmission-line model, which provides closed-form expressions for the initial dimensions of the radiating patch. These analytical relations establish a physically meaningful starting point for numerical optimization and enable the systematic evaluation of performance metrics such as directivity, gain, and voltage standing wave ratio. The antenna is designed to operate at the target frequency f_r on an FR-4 substrate characterized by a relative permittivity ϵ_r and thickness h . The free-space wavelength and the electromagnetic confinement within the dielectric jointly determine the effective electrical size of the patch. Consequently, both the physical dimensions and the substrate properties directly influence radiation efficiency, impedance matching, and bandwidth.

2.1. Directivity

Directivity quantifies the ability of an antenna to concentrate radiated power in a preferred spatial direction. It is defined as the ratio of the radiation intensity in a given direction to the average radiation intensity over all directions [37]. High directivity implies that electromagnetic energy is focused into a narrow beam, which is advantageous for RF energy harvesting and wireless power transfer, as it enhances the received power density in the intended direction. For microstrip antennas, directivity is primarily governed by the patch geometry, substrate characteristics, and surface current distribution on the radiating element.

2.2. Gain

Gain represents the combined effect of directivity and radiation efficiency [38]. It expresses the power radiated in a specific direction relative to that of an isotropic radiator with the same input power. Mathematically, gain is obtained by multiplying the radiation intensity in the direction of interest by 4π and normalizing it with respect to the total accepted input power. Unlike directivity, gain inherently

accounts for conduction and dielectric losses. In RF energy harvesting applications, gain is a critical parameter, as it directly influences the amount of power captured from ambient electromagnetic fields.

2.3. Voltage Standing Wave Ratio and Dimensional Formulation

The voltage standing wave ratio (VSWR) characterizes the impedance matching between the antenna and the feeding transmission line [6]. It is defined as the ratio of the maximum to minimum voltage amplitudes along the line. A VSWR value close to unity indicates near-perfect matching, whereas large values signify substantial reflections. In practical antenna systems, a VSWR below 2 is generally accepted as an indicator of satisfactory operation. For energy harvesting circuits, low VSWR is particularly important, as impedance mismatch directly reduces the power delivered to the rectification stage. For a rectangular microstrip patch antenna operating at frequency f_r on a substrate with dielectric constant ϵ_r , the width W of the patch is initially estimated as

$$W = \frac{c}{2f_r} \left(\frac{\epsilon_r + 1}{2} \right)^{-1/2} \quad (1)$$

where c denotes the speed of light in free space. This expression balances radiation efficiency and bandwidth by selecting a width that enhances fringing fields at the patch edges. Due to the partial propagation of electromagnetic fields in air and in the dielectric, the patch does not experience the full substrate permittivity. This effect is captured by the effective dielectric constant ϵ_{reff} , expressed as

$$\epsilon_{\text{reff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + \frac{12h}{W} \right)^{-1/2}, \text{ for } \frac{W}{h} > 1 \quad (2)$$

The finite thickness of the substrate causes fringing fields at the radiating edges, which electrically lengthen the patch. This phenomenon is modeled through an incremental length ΔL .

$$\Delta L = 0.412h \frac{(\epsilon_{\text{reff}} + 0.3) \left(\frac{W}{h} + 0.264 \right)}{(\epsilon_{\text{reff}} - 0.258) \left(\frac{W}{h} + 0.8 \right)} \quad (3)$$

The effective resonant length of the antenna becomes

$$L_{\text{eff}} = L + 2\Delta L = \frac{c}{2f_r \sqrt{\epsilon_{\text{reff}}}} \quad (4)$$

from which the physical patch length L is obtained as follows.

$$L = \frac{c}{2f_r \sqrt{\epsilon_{\text{reff}}}} - 2\Delta L = \frac{c}{2f_r \sqrt{\epsilon_{\text{reff}}}} - 0.824h \frac{(\epsilon_{\text{reff}} + 0.3) \left(\frac{W}{h} + 0.264 \right)}{(\epsilon_{\text{reff}} - 0.258) \left(\frac{W}{h} + 0.8 \right)} \quad (5)$$

In these expressions, L and W denote the physical length and width of the radiating patch, while h represents the substrate thickness. These relations provide a deterministic framework for the initial antenna geometry prior to numerical refinement. For circular microstrip antennas or circular radiating sections, the equivalent formulation is expressed through an auxiliary parameter F .

$$F = \frac{8.79 \times 10^9}{f_r \sqrt{\epsilon_r}} \quad (6)$$

and the effective radius r is obtained as follows.

$$r = \frac{F}{\left[1 + \frac{24}{\pi F \epsilon_r} \left(\ln \left(\frac{\pi F}{24}\right) + 1.7726\right)\right]^{1/2}} \quad (7)$$

These analytical relations ensure that the antenna resonates at the desired operating frequency while maintaining appropriate impedance characteristics. In the present work, the rectangular formulation serves as the foundation for the proposed geometries. The analytically obtained dimensions are subsequently refined through full-wave simulations and geometric modifications, enabling the achievement of low VSWR, reduced reflection coefficient, and enhanced gain, which are essential for efficient RF energy harvesting.

3. Simulation Results of Designed Antennas

RF energy harvesting relies on the efficient capture of weak ambient electromagnetic fields and their conversion into usable electrical energy. In such systems, antenna performance becomes the dominant factor governing the overall efficiency of the harvesting chain. A survey of recent microstrip antenna designs revealed several recurring limitations, including insufficient geometric miniaturization, modest gain levels, and inadequate suppression of reflected power. These drawbacks directly restrict the effectiveness of energy harvesting circuits, particularly under low-power excitation conditions. To address these issues, the present study adopts a geometry-driven enhancement strategy. Unique slots and notches are introduced on the radiating surface in order to reshape the surface current distribution, extend the effective electrical path, and improve impedance matching. The design process is conducted iteratively in CST Microwave Studio, where each geometric modification is evaluated in terms of reflection coefficient, VSWR, gain, and radiation characteristics. The objective is to satisfy the fundamental criteria for RF energy harvesting antennas, namely:

- VSWR ≤ 2 , ensuring proper impedance matching,
- $S_{11} < -10$ dB, indicating minimal reflected power [39],
- Adequate gain and directivity for efficient power capture.

Symmetrical notches are therefore positioned on both sides of the feed line and oriented toward the lower edges of the radiating surface, following principles reported in earlier bandwidth-enhancement studies. The resulting geometries are evaluated through hundreds of full-wave simulations, leading to two optimized antenna configurations: an initial reference design and a compact, improved structure.

3.1. First Antenna Design: Parametric Exploration and Baseline Performance

The geometry of the first antenna is illustrated in Fig. 1. For systematic optimization, the antenna dimensions are parameterized as follows: the width and length of the radiating surface are denoted by x_{ant} and y_{ant} , respectively; the ground plane and dielectric substrate share equal dimensions, represented by ntx and nty ; and the width and length of the slots and notches are defined as $oyukx$ and $oyuky$.

Approximately 620 simulation runs are performed by varying these parameters within physically meaningful bounds. From this extensive design space, the ten configurations exhibiting the most favorable electromagnetic performance are selected. Their dimensions are summarized in Table 1. The reflection characteristics of these candidates are presented in Fig. 2. The resonance occurs around 2.4 GHz for all selected configurations, with reflection coefficients ranging from -27.21 dB to -29.74 dB. The operational bandwidth spans approximately 2.358–2.44 GHz, fully covering the target ISM band. The minimum S_{11} value of -29.74 dB at 2.4 GHz confirms highly effective impedance matching and minimal power loss due to reflection. Impedance behavior is further evaluated through the VSWR curves shown in Fig. 3. At the resonance frequency, all configurations exhibit VSWR values between 1.0 and 1.09, comfortably satisfying the $VSWR \leq 2$ requirement. These results indicate near-ideal matching between the antenna and the feeding line.

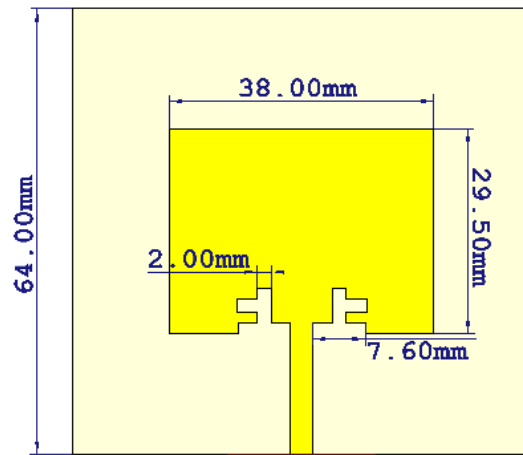


Fig. 1. First designed microstrip patch antenna

Table 1. Dimensions of the simulated initial antennas

No	xant	yant	ntx	nty	oyukx	oyuky
111	37.5	29.5	58	62	7.4	1.6
121	37.5	29.5	58	62.2	7.4	1.6
122	37.5	29.5	60	62.2	7.4	1.6
123	37.5	29.5	62	62.2	7.4	1.6
131	37.5	29.5	58	62.4	7.4	1.6
132	37.5	29.5	60	62.4	7.4	1.6
141	37.5	29.5	58	62.6	7.4	1.6
143	37.5	29.5	62	62.6	7.4	1.6
151	37.5	29.5	58	62.8	7.4	1.6
161	37.5	29.5	58	63	7.4	1.6

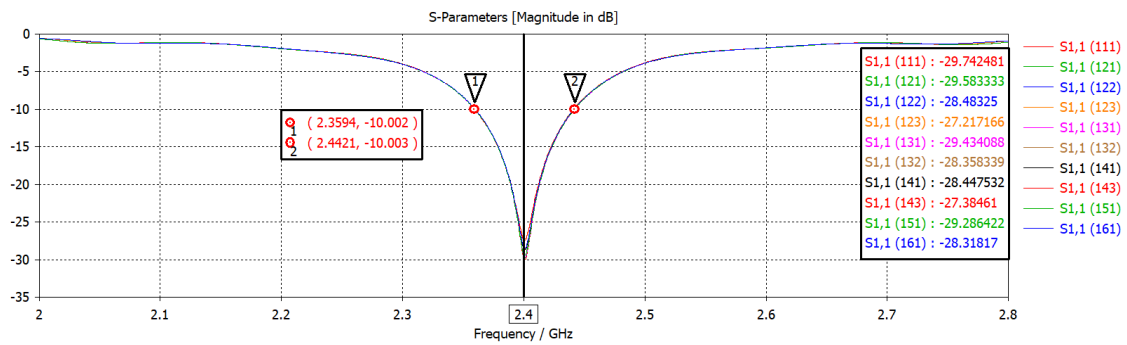
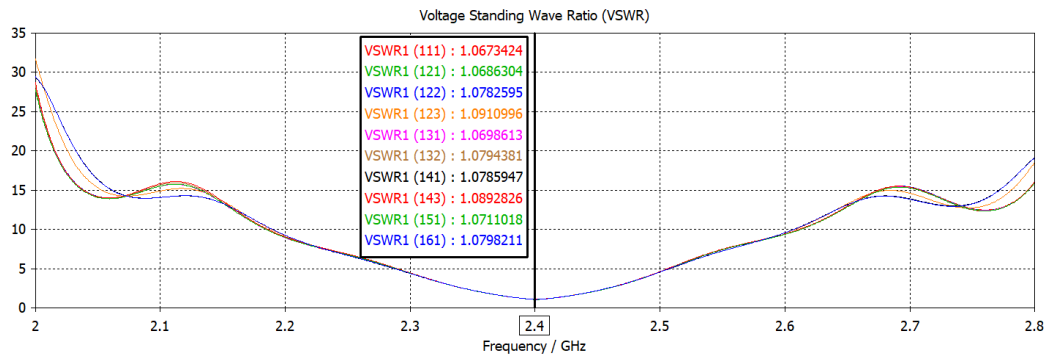
Fig. 2. S₁₁ Graph of the first designed antennas

Fig. 3. VSWR Graph of the first designed antennas

Radiation characteristics are examined using both one-dimensional and three-dimensional patterns. The one-dimensional gain and directivity responses are illustrated in Fig. 4, while their spatial distributions are shown in Fig. 5. At 2.4 GHz, the antenna exhibits a peak directivity of 6.6 dBi and a gain of 2.67 dBi. The main lobe is well-defined, and the side-lobe level is approximately -14.7 dBi, indicating controlled radiation behavior. The angular beamwidth of 93.5° and the narrow beam profile confirm suitability for directional energy capture. Collectively, these results demonstrate that the first antenna configuration satisfies the performance requirements prescribed by IEEE 802.11b standards.

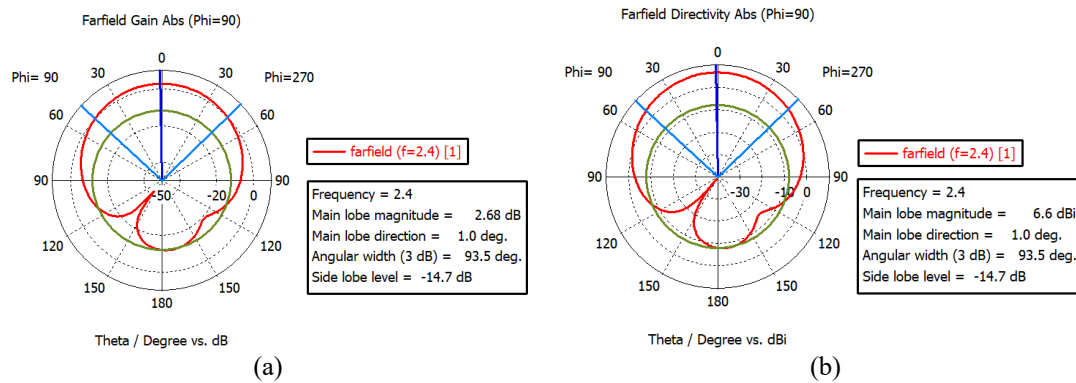


Fig. 4. One-dimensional gain graph of the first designed antenna (a) and one-dimensional directivity graph (b) of the first designed antenna

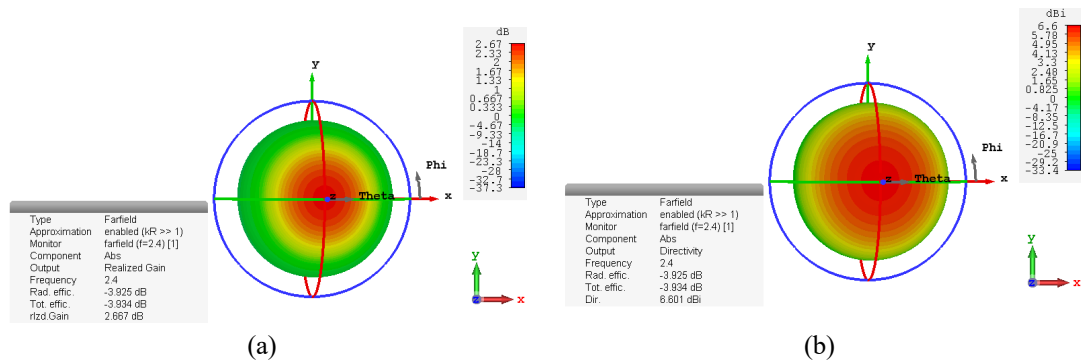


Fig. 5. Three-dimensional gain graph of the first designed antenna (a) and three-dimensional directivity graph (b) of the first designed antenna

3.2. Second Antenna Design: Compact Geometry and Enhanced Performance

Building upon the baseline design, further geometric refinements are applied to reduce physical size while preserving or enhancing electromagnetic performance. The modified structure incorporates revised slot dimensions and a compact substrate outline. The resulting parameter sets obtained from iterative simulations are summarized in Table 2.

Table 2. Dimensions of the simulated second antenna

No	Xant	yant	Sbx	Sby	oyukx	oyuky
248	38	29	50	40	6	2
367	37.5	29	40	40	6.5	2
392	38	29	50	40	7	1.5
415	37.5	29	40	40	7	1.5
489	38.5	29	50	40	7.5	1.5
513	38.5	29	40	40	7.5	1.5

The finalized geometry is depicted in Fig. 6. The designed and fabricated microstrip patch antenna, and the fabricated prototype is shown in Fig. 7. Compared with the first design, the second antenna achieves noticeable miniaturization without compromising impedance or radiation characteristics.

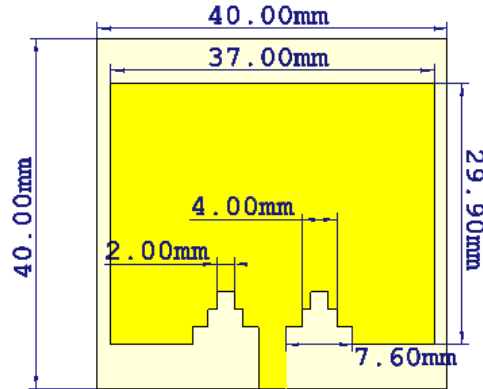


Fig. 6. The designed and fabricated microstrip patch antenna

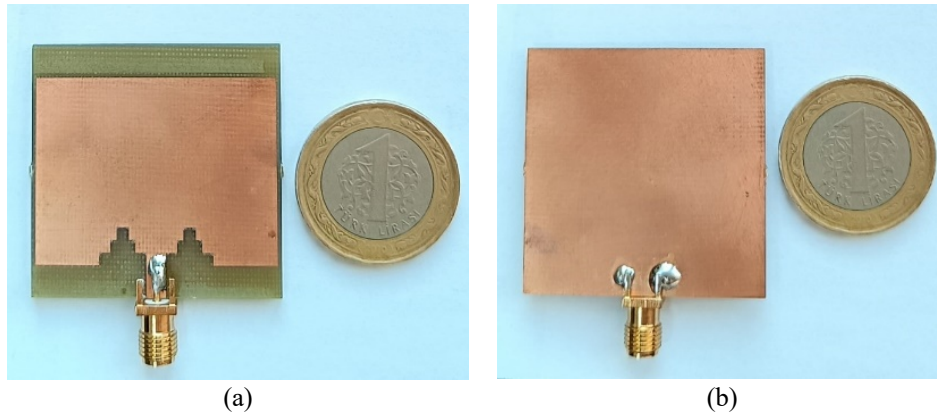


Fig. 7. Front (a) and rear (b) views of the designed and fabricated second (single band) antenna

The reflection coefficient of the second antenna is illustrated in Fig. 8. Across the operational band of 2.35–2.44 GHz, S_{11} values remain well below -10 dB, reaching a minimum of -31.01 dB at 2.4 GHz. This behavior confirms precise tuning and superior impedance matching. Corresponding VSWR characteristics are presented in Fig. 9. The VSWR equals unity at the resonance frequency and remains below 2 throughout the operating band, indicating excellent compatibility with the feeding network.

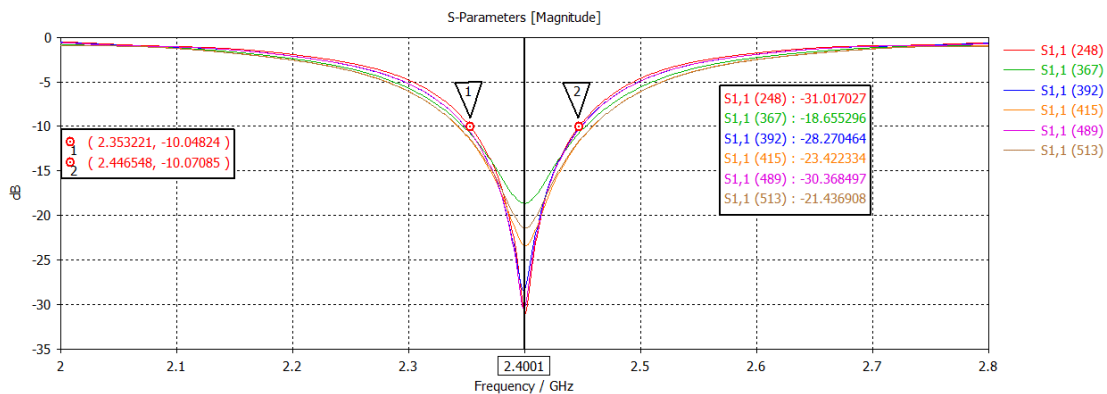


Fig. 8. S_{11} Graphs of the designed second antenna

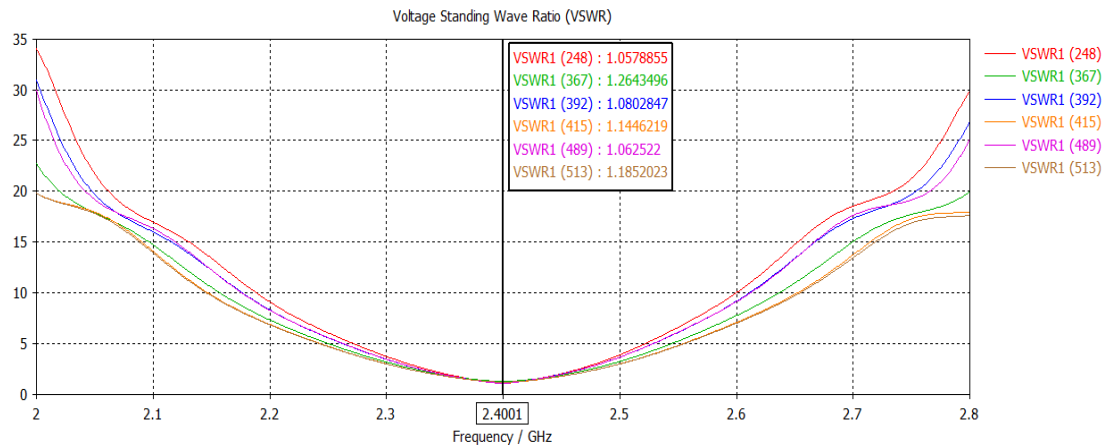


Fig. 9. VSWR graphs of the designed second antenna

Radiation properties are analyzed using the one-dimensional patterns in Fig. 10. The main lobe is oriented at approximately 6° , with an angular beamwidth of 98.0° and a peak gain of 2.75 dBi. The side-lobe level is measured at -7.3 dBi, reflecting a balanced radiation profile. Three-dimensional radiation distributions are shown in Fig. 11. The directivity reaches 5.08 dBi, while the gain remains at 2.75 dBi. These results confirm that the geometric modifications effectively improve performance while maintaining compactness. The antenna parameters comply with IEEE 802.11b/g specifications in terms of frequency coverage, impedance behavior, and radiation efficiency.

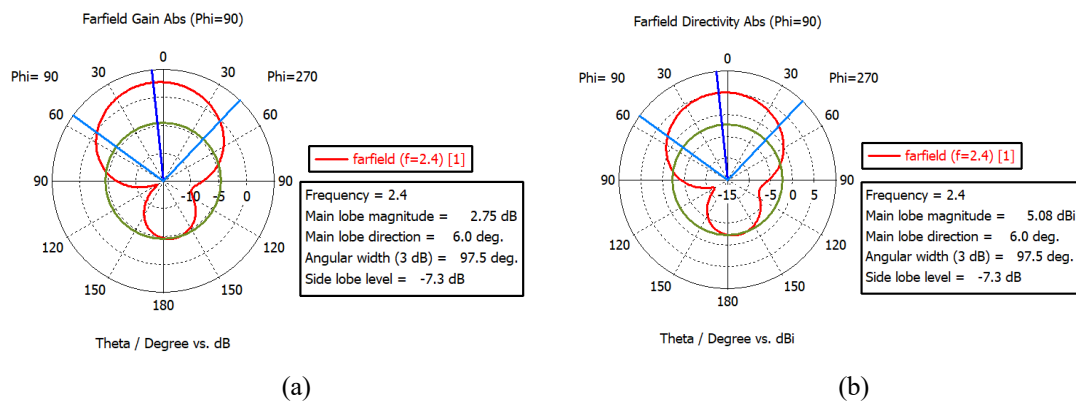


Fig. 10. One-dimensional gain graph (a) and one-dimensional directivity graph (b) of the designed second antenna

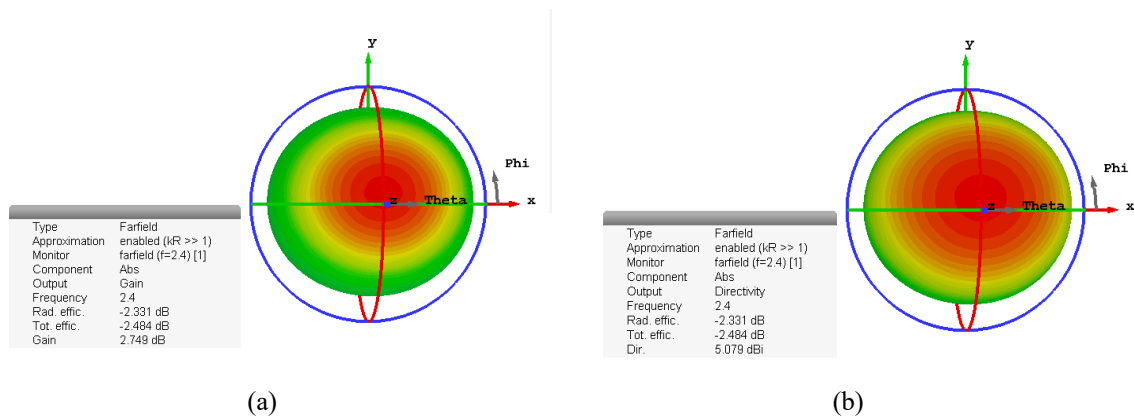


Fig. 11. Three-dimensional gain graph (a) and three-dimensional directivity graph (b) of the designed second antenna

3.3. Experimental Validation and Energy Harvesting Assessment

The fabricated antenna is evaluated using a vector network analyzer, and the measured response is presented in Fig. 12. The measured S_{11} curve closely follows the simulated response, confirming the reliability of the design methodology and the accuracy of the fabrication process. To assess practical energy harvesting capability, the antenna is connected to a Powercast P21XXCSR-EVB energy harvesting module, shown in Fig. 13. The electrical characteristics of this module are summarized in Table 3. The module performs rectification, voltage regulation, and storage, enabling the conversion of captured RF power into usable DC output.



Fig. 12. Network analyzer results of the designed second antenna

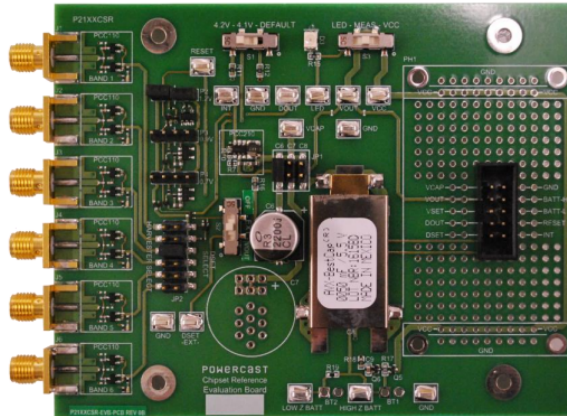


Fig. 13. P21XXCSR-EVB energy harvesting module

Table 3. Parameters of the P21XXCSR-EVB energy harvesting module

Parameter	Degree	Unit
RF input power	23	dBm
RF _{IN} to GND	0	V
D _{set} to GND	6	V
Reset to GND	6	V
V _{CAP} to GND	2.3	V
V _{OUT} to GND	6	V
V _{OUT} TO current	100	mA
Operating temperature range	-40 to 85	°C
Storage temperature range	-40 to 140	°C

The harvested voltage is measured using a true RMS multimeter, and the result is presented in Fig. 14. Voltage measurement result of the designed second antenna at 2.4 GHz. Under operating conditions, the antenna produces a DC voltage of 3.51 V. This value confirms that the proposed compact single-band antenna is capable of delivering sufficient power for low-energy electronic systems [40]. Overall, the simulation and experimental results demonstrate that the proposed design strategy (based on analytical formulation, parametric simulation, and geometry-driven refinement) yields a compact, well-matched, and efficient antenna suitable for RF energy harvesting and wireless power transfer applications.



Fig. 14. Voltage measurement result of the designed second antenna at 2.4 GHz

4. Conclusion

In this work, two single-band rectangular microstrip patch antennas operating at 2.4 GHz were developed for RF energy harvesting applications. The design procedure was initiated using analytical formulations derived from the transmission-line model, providing physically meaningful starting dimensions for the radiating patch and substrate. These initial geometries were subsequently refined through extensive full-wave simulations in CST Microwave Studio, where geometric parameters were systematically adjusted to improve impedance matching, radiation behavior, and compactness. A geometry-driven enhancement strategy was adopted in which unique slots and notches were introduced on the radiating surface. These structural modifications reshaped surface current paths effectively extended the electrical length of the patch, and enabled improved control over bandwidth and resonance. Hundreds of parametric simulations were conducted, allowing the progressive evolution of the antenna geometry toward optimal performance. The influence of each modification on bandwidth, operating frequency, directivity, VSWR, reflection coefficient, and gain was evaluated through detailed graphical analysis, revealing consistent and measurable improvements. Following the identification of the optimal configuration, one antenna was fabricated on an FR-4 substrate ($\epsilon_r = 4.4$) using standard printed circuit board technology. Experimental validation was carried out using a vector network analyzer, and the measured impedance characteristics were found to be in close agreement with the simulation results. The fabricated antenna achieved a resonance at 2.4 GHz with a VSWR of 1.0 and a reflection coefficient of -31.01 dB, confirming excellent impedance matching.

The measured gain and directivity reached 2.75 dBi and 5.08 dBi, respectively, demonstrating that compactness was attained without sacrificing radiation efficiency. To assess practical applicability, the antenna was integrated with a commercial RF energy harvesting module. The experimental setup included a network analyzer, a horn antenna, and a True RMS multimeter. Under operating conditions, the fabricated antenna delivered a DC voltage of 3.51 V, verifying its ability to supply usable energy to low-power electronic systems. These results demonstrate that the proposed design methodology—combining analytical formulation, geometry-driven refinement, and experimental verification—yields a compact, low-cost, and well-matched antenna suitable for RF energy harvesting and wireless power transfer. The presented approach offers a reproducible pathway for the development of efficient harvesting antennas and can be readily extended to other frequency bands and emerging low-power wireless platforms.

Author Contribution: All authors contributed equally to the main contributor to this paper. All authors read and approved the final paper.

Funding: This research received no external funding.

Conflicts of Interest: The authors declare no conflict of interest.

References

- [1] H. Özmen, D. Izci, R. M. Rizk-Allah, S. Ekinici, and M. Dalarsson, "Optimization of split-ring resonator slots using levy-opposition-enhanced Newton–Raphson method for high-gain UWB Vivaldi antenna design," *Scientific Reports*, vol. 16, no. 1, p. 7828, Feb. 2026, <https://doi.org/10.1038/s41598-026-41244-5>.
- [2] R. Singh and R. Kaur, "A novel Archimedes optimization algorithm with levy flight for designing microstrip patch antenna," *Arabian Journal for Science and Engineering*, vol. 47, no. 3, pp. 3683–3706, Mar. 2022, <https://doi.org/10.1007/s13369-021-06307-x>.
- [3] P. Moeikham and P. Akkaraekthalin, "A compact printed slot antenna with high out-of-band rejection for WLAN/WiMAX applications," *Radioengineering*, vol. 25, no. 4, pp. 672–679, Dec. 2016, <https://doi.org/10.13164/re.2016.0672>.
- [4] S. K. Ezzulddin, S. O. Hasan, and M. M. Ameen, "Microstrip patch antenna design, simulation and fabrication for 5G applications," *Simulation Modelling Practice and Theory*, vol. 116, p. 102497, Apr. 2022, <https://doi.org/10.1016/j.simpat.2022.102497>.
- [5] M. S. Rana, S. Hossain, S. B. Rana, and M. M. Rahman, "Microstrip patch antennas for various applications: A review," *Indonesian Journal of Electrical Engineering and Computer Science*, vol. 29, no. 3, pp. 1511–1519, Mar. 2023, <https://doi.org/10.11591/ijeecs.v29.i3.pp1511-1519>.
- [6] C. A. Balanis, *Antenna theory: Analysis and design*, 3rd ed. Hoboken, NJ, USA: John Wiley & Sons, 2005, <https://books.google.co.id/books?id=jwjeAwAAQBAJ>.
- [7] G. Misra, A. Agarwal, and K. Agarwal, "Design and performance evaluation of microstrip antenna for ultra-wideband applications using microstrip feed," *American Journal of Electrical and Electronic Engineering*, vol. 3, no. 4, pp. 93–99, 2015, <http://pubs.sciepub.com/ajeec/3/4/2>.
- [8] H. Özmen, S. Ekinici, and D. Izci, "Boosted arithmetic optimization algorithm with elite opposition-based pattern search mechanism and its promise to design microstrip patch antenna for WLAN and WiMAX," *International Journal of Modelling and Simulation*, vol. 45, no. 1, pp. 1–16, Mar. 2023, <https://doi.org/10.1080/02286203.2023.2196736>.
- [9] C. Benkalfate, A. Ouslimani, A.-E. Kasbari, and M. Feham, "A new compact triple-band triangular patch antenna for RF energy harvesting applications in IoT devices," *Sensors*, vol. 22, no. 20, p. 8009, Oct. 2022, <https://doi.org/10.3390/s22208009>.

- [10] D. Kumar, C. Anushka, S. D. Choudhary, and V. Shekher, "Design and simulation of I-slot microstrip patch antenna with defected ground structure for 5G applications," in *Proceedings of the 2018 2nd International Conference on Micro-Electronics and Telecommunication Engineering (ICMETE)*, Ghaziabad, India, Sep. 2018, pp. 187–190, <https://doi.org/10.1109/ICMETE.2018.00049>.
- [11] J. Qiao, X. Shen, J. W. Mark, Q. Shen, Y. He, and L. Lei, "Enabling device-to-device communications in millimeter-wave 5G cellular networks," *IEEE Communications Magazine*, vol. 53, no. 1, pp. 209–215, Jan. 2015, <https://doi.org/10.1109/MCOM.2015.7010536>.
- [12] A. Mehdipour, A. Sebak, C. W. Trueman, and T. A. Denidni, "Compact multiband planar antenna for 2.4/3.5/5.2/5.8-GHz wireless applications," *IEEE Antennas and Wireless Propagation Letters*, vol. 11, pp. 144–147, 2012, <https://doi.org/10.1109/LAWP.2012.2185915>.
- [13] H. Sun, "An enhanced rectenna using differentially-fed rectifier for wireless power transmission," *IEEE Antennas and Wireless Propagation Letters*, vol. 15, pp. 32–35, 2016, <https://doi.org/10.1109/LAWP.2015.2427197>.
- [14] U. Olgun, C.-C. Chen, and J. L. Volakis, "Wireless power harvesting with planar rectennas for 2.45 GHz RFIDs," in *Proceedings of the 2010 URSI International Symposium on Electromagnetic Theory*, Berlin, Germany, Aug. 2010, pp. 329–331, <https://doi.org/10.1109/URSI-EMTS.2010.5637008>.
- [15] G. Vellidis, M. Tucker, C. Perry, C. Kvien, and C. Bednarz, "A real-time wireless smart sensor array for scheduling irrigation," *Computers and Electronics in Agriculture*, vol. 61, no. 1, pp. 44–50, Apr. 2008, <https://doi.org/10.1016/j.compag.2007.05.009>.
- [16] H. Malekpoor and S. Jam, "Design of a multi-band asymmetric patch antenna for wireless applications," *Microwave and Optical Technology Letters*, vol. 55, no. 4, pp. 730–734, Apr. 2013, <https://doi.org/10.1002/mop.27449>.
- [17] S. Sahoo, L. P. Mishra, and M. N. Mohanty, "Optimization of Z-shape microstrip antenna with I-slot using discrete particle swarm optimization (DPSO) algorithm," *Procedia Computer Science*, vol. 92, pp. 91–98, 2016, <https://doi.org/10.1016/j.procs.2016.07.328>.
- [18] S. Dey, A. Sinha, B. Gill, M. Gangopadhyaya, and S. Ray, "Resonant frequency optimization of rectangular gap coupled microstrip antenna using particle swarm optimization algorithm," in *Proceedings of the 2016 IEEE 7th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON)*, Vancouver, BC, Canada, Oct. 2016, pp. 1–4, <https://doi.org/10.1109/IEMCON.2016.7746360>.
- [19] B. Manoj and S. Rodriguez, "Genetic algorithm optimization of microstrip patch antenna dimensions for enhanced bandwidth, accurate operating frequency and return loss," in *Proceedings of the 2021 Sixth International Conference on Wireless Communications, Signal Processing and Networking (WiSPNET)*, Chennai, India, Mar. 2021, pp. 441–445, <https://doi.org/10.1109/WiSPNET51692.2021.9419477>.
- [20] M. A. Motin, M. I. Hasan, and M. Asaduz-Zaman, "Design and optimization of a low cost multi band microstrip patch antenna for K-band, Ku-band and X-band applications," in *Proceedings of the 2012 15th International Conference on Computer and Information Technology (ICCIT)*, Chittagong, Bangladesh, Dec. 2012, pp. 615–620, <https://doi.org/10.1109/ICCITech.2012.6509750>.
- [21] H. Özmen and M. B. Kurt, "Radar-based microwave breast cancer detection system with a high-performance ultrawide band antipodal Vivaldi antenna," *Turkish Journal of Electrical Engineering and Computer Sciences*, vol. 29, no. 5, pp. 2326–2345, Sep. 2021, <https://doi.org/10.3906/elk-2010-49>.
- [22] S. Imtiaj and S. Bhattacharya, "Performance comparison between 2×1 and 2×2 corporate feed antenna array in X-band," in *Proceedings of the 2020 IEEE Calcutta Conference (CALCON)*, Kolkata, India, Feb. 2020, pp. 189–193, <https://doi.org/10.1109/CALCON49167.2020.9106492>.
- [23] J. M. J. W. Jayasinghe, J. Anguera, D. N. Uduwawala, and A. Andújar, "Nonuniform overlapping method in designing microstrip patch antennas using genetic algorithm optimization," *International Journal of Antennas and Propagation*, vol. 2015, pp. 1–8, 2015, <https://doi.org/10.1155/2015/805820>.
- [24] H. M. Q. Rasheda, N. M. Shah, A. Saif, Q. Abdullah, A. Uğurenver, A. R. O. Mumin, and N. B. M. Sahar, "An optimization of fractal microstrip patch antenna with partial ground using genetic algorithm method,"

- in *Proceedings of the 2021 International Congress of Advanced Technology and Engineering (ICOTEN)*, Taiz, Yemen, Jul. 2021, pp. 1–6, <https://doi.org/10.1109/ICOTEN52080.2021.9493531>.
- [25] P. Zhang, H. Yi, H. Liu, H. Yang, G. Zhou, and L. Li, "Back-to-back microstrip antenna design for broadband wide-angle RF energy harvesting and dedicated wireless power transfer," *IEEE Access*, vol. 8, pp. 126868–126875, 2020, <https://doi.org/10.1109/ACCESS.2020.3008551>.
- [26] T. Orankitanun and S. Yaowiwat, "Application of genetic algorithm in tri-band U-slot microstrip antenna design," in *Proceedings of the 2020 17th International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology (ECTI-CON)*, Phuket, Thailand, Jun. 2020, pp. 127–130, <https://doi.org/10.1109/ECTI-CON49241.2020.9158066>.
- [27] M. Boudjerda, A. Reddaf, A. Kacha, K. Hamdi-Cherif, T. E. A. Alharbi, M. S. Alzaidi, M. Alsharef, and S. S. M. Ghoneim, "Design and optimization of miniaturized microstrip patch antennas using a genetic algorithm," *Electronics*, vol. 11, no. 14, p. 2123, 2022, <https://doi.org/10.3390/electronics11142123>.
- [28] M. T. Islam, M. Z. Mahmud, M. T. Islam, S. Kibria, and M. Samsuzzaman, "A low cost and portable microwave imaging system for breast tumor detection using UWB directional antenna array," *Scientific Reports*, vol. 9, no. 1, p. 15491, Oct. 2019, <https://doi.org/10.1038/s41598-019-51620-z>.
- [29] M. S. H. Mollah, O. Faruk, M. S. Hossain, M. T. Islam, A. S. M. Shafi, and M. M. I. Molla, "Design and performance improvement of microstrip patch antenna using graphene material for communication applications," in *Proceedings of the 2021 IEEE 11th Symposium on Computer Applications and Industrial Electronics (ISCAIE)*, Penang, Malaysia, Apr. 2021, pp. 343–346, <https://doi.org/10.1109/ISCAIE51753.2021.9431793>.
- [30] T. Yathavi, K. Maunasree, G. B. Meenakshi, M. V. Malika, and M. Santhoshini, "RF energy harvesting for low power applications," in *Proceedings of the 2021 10th International Conference on Internet of Everything, Microwave Engineering, Communication and Networks (IEMECON)*, Jaipur, India, Dec. 2021, pp. 1–6, <https://doi.org/10.1109/IEMECON53809.2021.9689155>.
- [31] M. Cansiz, D. Altinel, and G. K. Kurt, "Efficiency in RF energy harvesting systems: A comprehensive review," *Energy*, vol. 174, pp. 292–309, May 2019, <https://doi.org/10.1016/j.energy.2019.02.100>.
- [32] J. Zhang, Z. P. Wu, C. G. Liu, B. H. Zhang, and B. Zhang, "A double-sided rectenna design for RF energy harvesting," in *Proceedings of the 2015 IEEE International Wireless Symposium (IWS 2015)*, Shenzhen, China, Mar. 2015, pp. 1–4, <https://doi.org/10.1109/IEEE-IWS.2015.7164617>.
- [33] X. Qi, Z. Xu, and H. Li, "High-efficiency 2-D multibeam rectenna based on gain enhanced patch array," *IEEE Antennas and Wireless Propagation Letters*, vol. 21, no. 12, pp. 2537–2541, Dec. 2022, <https://doi.org/10.1109/LAWP.2022.3200516>.
- [34] H. Chen, L. Zhang, Y. Chen, C. Mao, W. Li, and Y. He, "Reverse design of a dual-band microstrip patch antenna based on binary particle swarm optimization," in *Proceedings of the 2023 International Applied Computational Electromagnetics Society Symposium (ACES-China)*, Hangzhou, China, Aug. 2023, pp. 1–3, <https://doi.org/10.23919/ACES-China60289.2023.10249454>.
- [35] P. R. Prajapati, G. G. K. Murthy, A. Patnaik, and M. V. Kartikeyan, "Design and testing of a compact circularly polarised microstrip antenna with fractal defected ground structure for L-band applications," *IET Microwaves, Antennas & Propagation*, vol. 9, no. 11, pp. 1179–1185, Aug. 2015, <https://doi.org/10.1049/iet-map.2014.0596>.
- [36] M. T. Islam, M. Z. Mahmud, N. Misran, J.-I. Takada, and M. Cho, "Microwave breast phantom measurement system with compact side slotted directional antenna," *IEEE Access*, vol. 5, pp. 5321–5330, 2017, <https://doi.org/10.1109/ACCESS.2017.2690671>.
- [37] F. Mahbub, R. Islam, S. A. Kadir Al-Nahyun, S. B. Akash, R. R. Hasan, and M. A. Rahman, "A single-band 28.5 GHz rectangular microstrip patch antenna for 5G communications technology," in *Proceedings of the 2021 IEEE 11th Annual Computing and Communication Workshop and Conference (CCWC)*, Las Vegas, NV, USA, Jan. 2021, pp. 1151–1156, <https://doi.org/10.1109/CCWC51732.2021.9376047>.

-
- [38] S. K. Ibrahim and Z. T. Jebur, "A high gain compact rectangular patch antenna for 5G applications," in *Proceedings of the 2021 International Conference on Communication and Information Technology (ICICT)*, Basrah, Iraq, Jun. 2021, pp. 156–160, <https://doi.org/10.1109/ICICT52195.2021.9568415>.
- [39] M. K. Shereen, M. I. Khattak, and J. Nebhen, "A review of achieving frequency reconfiguration through switching in microstrip patch antennas for future 5G applications," *Alexandria Engineering Journal*, vol. 61, no. 1, pp. 29–40, Jan. 2022, <https://doi.org/10.1016/j.aej.2021.04.105>.
- [40] S. Gautam, S. Kumar, S. Chatzinotas, and B. Ottersten, "Experimental evaluation of RF waveform designs for wireless power transfer using software defined radio," *IEEE Access*, vol. 9, pp. 132609–132622, 2021, <https://doi.org/10.1109/ACCESS.2021.3115048>.